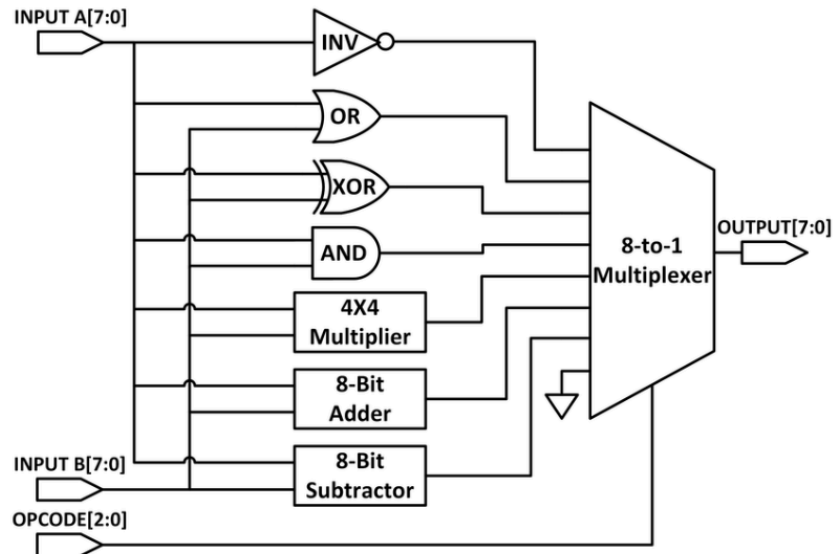
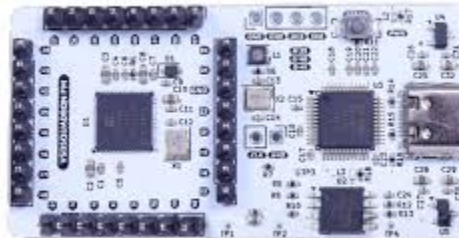


1. Design and implement an 8-bit ALU on FPGA with inputs A[7:0], B[7:0], and control signal OPCODE[2:0]. The ALU shall support logical operations (NOT, AND, OR, XOR), arithmetic operations (8-bit addition and subtraction), and a 4×4 multiplication. Use an 8-to-1 multiplexer to select the final output based on the opcode. Implement the design in Verilog and verify its functionality through simulation.



IMPLIMENTED ON VSD SQUADRON FM



CODE:

```
module ALU (  
input [2:0] opcode ,  
output [7:0] out );  
  
wire clk ;  
SB_HFOSC osc_int(  
    .CLKHFPU(1'b1),  
    .CLKHFEN(1'b1),  
    .CLKHF(clk)  
);  
defparam osc_int.CLKHF_DIV = "0b10"; //12MHZ  
  
reg [7:0] a = 8'b10101010 ;  
reg [7:0] b = 8'b11110000 ;  
  
always @(posedge clk ) begin  
case (opcode )  
    3'b000: out <= ~a ;  
    3'b001: out <= a | b ;  
    3'b010: out <= a ^ b ;  
    3'b011: out <= a & b ;  
    3'b100: out <= a * b ;  
    3'b101: out <= a + b ;  
    3'b110: out <= a - b ;  
  
    default : out <= a ^ b ; //default for xor case  
endcase  
end  
endmodule
```

CONS:

```
set_io opcode[0] 37  
set_io opcode[1] 36  
set_io opcode[2] 35  
set_io out[0] 3  
set_io out[1] 4  
set_io out[2] 6  
set_io out[3] 9  
set_io out[4] 10  
set_io out[5] 11  
set_io out[6] 12  
set_io out[7] 39
```

SYNTH REPORTS

```
2.44. Printing statistics.

=== ALU ===

Number of wires:          12
Number of wire bits:      35
Number of public wires:   5
Number of public wire bits: 28
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          14
  SB_DFFSR                 3
  SB_DFFSS                 3
  SB_HFOSC                 1
  SB_LUT4                  7

2.45. Executing CHECK pass (checking for obvious problems).
checking module ALU..
found and reported 0 problems.

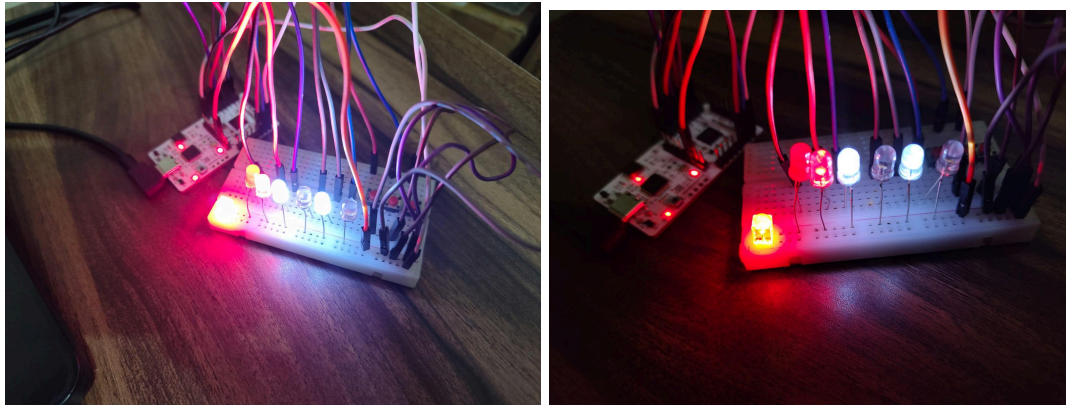
2.46. Executing JSON backend.

Warnings: 8 unique messages, 8 total
End of script. Logfile hash: 9a2e4ca31f
CPU: user 0.36s system 0.04s, MEM: 45.71 MB total, 40.20 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: 46% 10x read_verilog (0 sec), 13% 1x share (0 sec), ...
```

IMPLIMENTATION

```
Info: Program finished normally.
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_44$ icepack alu.asc alu.bin
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_44$ iceprog alu.bin
init..
cdone: high
reset..
cdone: low
flash ID: 0xEF 0x40 0x16 0x00
file size: 104090
erase 64kB sector at 0x000000..
erase 64kB sector at 0x010000..
programming..
done.
reading..
VERIFY OK
cdone: high
Bye.
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_44$
```

OUTPUTS;



VIDEO:https://github.com/KPYerawar/NIELIT-CALICUT-FPGA-PROTOTYPING-ASSIGNMENTS/blob/main/EXP_NO_44/video1.mp4

GITHUB:https://github.com/KPYerawar/NIELIT-CALICUT-FPGA-PROTOTYPING-ASSIGNMENTS/tree/main/EXP_NO_44